

Introduction to Operating Systems

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Outline:

- ▶ Course, Reference Books
- ▶ Why to learn Operating Systems?
- ▶ What is an Operating System?
- ▶ Functions of Operating Systems
- ▶ Layers of Operating System
- ▶ Types of Operating Systems

CS 3009

- ▶ Attendance: As per NIT rules.
- ▶ **2 Assignments and 2 Surprise Tests.**
- ▶ **Any kind of Copy/Plagiarism lead to `F` Grade**

CS 3009 - Text Books

- ▶ **Operating System Concepts:** by Abraham Silberschatz, Peter Galvin, Gagne, 9th Edition, 2016.
- ▶ **Modern Operating Systems:** by Andrew S. Tanenbaum, Herbert Bos, 4th Edition
- ▶ **Operating Systems: Internals and Design Principles:** by William Stallings, 8th Edition

Objective

- ▶ General understanding of structure of modern computers
- ▶ To understand the purpose, structure and functions of operating systems
- ▶ To illustrate key OS aspects by example
- ▶ To study and apply concepts relating to operating systems, such as concurrency, deadlocks, memory management, processor and disk scheduling, parallel processing, and file system organization

Outcome

- ▶ Explain the basic structure and functioning of OS
- ▶ Differentiate between processes and threads
- ▶ Point the problems related to process management and synchronization and apply learned methods to solve basic problems
- ▶ Explain the cause and effect related to deadlocks and to analyse them related to common circumstances in operating systems
- ▶ Explain the basics of memory management, the use of virtual memory in modern operating systems and the structure of the most common file-systems

What is an OS?

- ▶ Consider a bare machine with hardware only.
 - ▶ Can we write a simple program and execute it on h/w directly; without any s/w?
 - ▶ **YES**
 - ▶ Why do we need software's then?
 - ▶ **Convenience**
- ▶ Operating System (OS) is a collection of software's that manages hardware's.
- ▶ OS is an *interface* between the user and the hardware.

What is an OS?

- ▶ **Interface:** between Application Programs (USERs) and the hardware.
- ▶ **Server:** Provides services.
- ▶ **Resource Manager:**
 - ▶ CPU: OS assigns CPU to different tasks being executed.
 - ▶ Main memory: Processes are to be stored.
 - ▶ **Secondary Memory:** Programs/Files are to be stored.
 - ▶ **I/O devices:** Co-ordination and assignment of different I/O devices among the executing processes.

Abstract view of a Computer System

- ▶ 2 view points of OS

- ▶ USER View (Convenience)

- ▶ SYSTEM View (many complicated tasks that are abstracted from users)

Abstract view of a Computer System

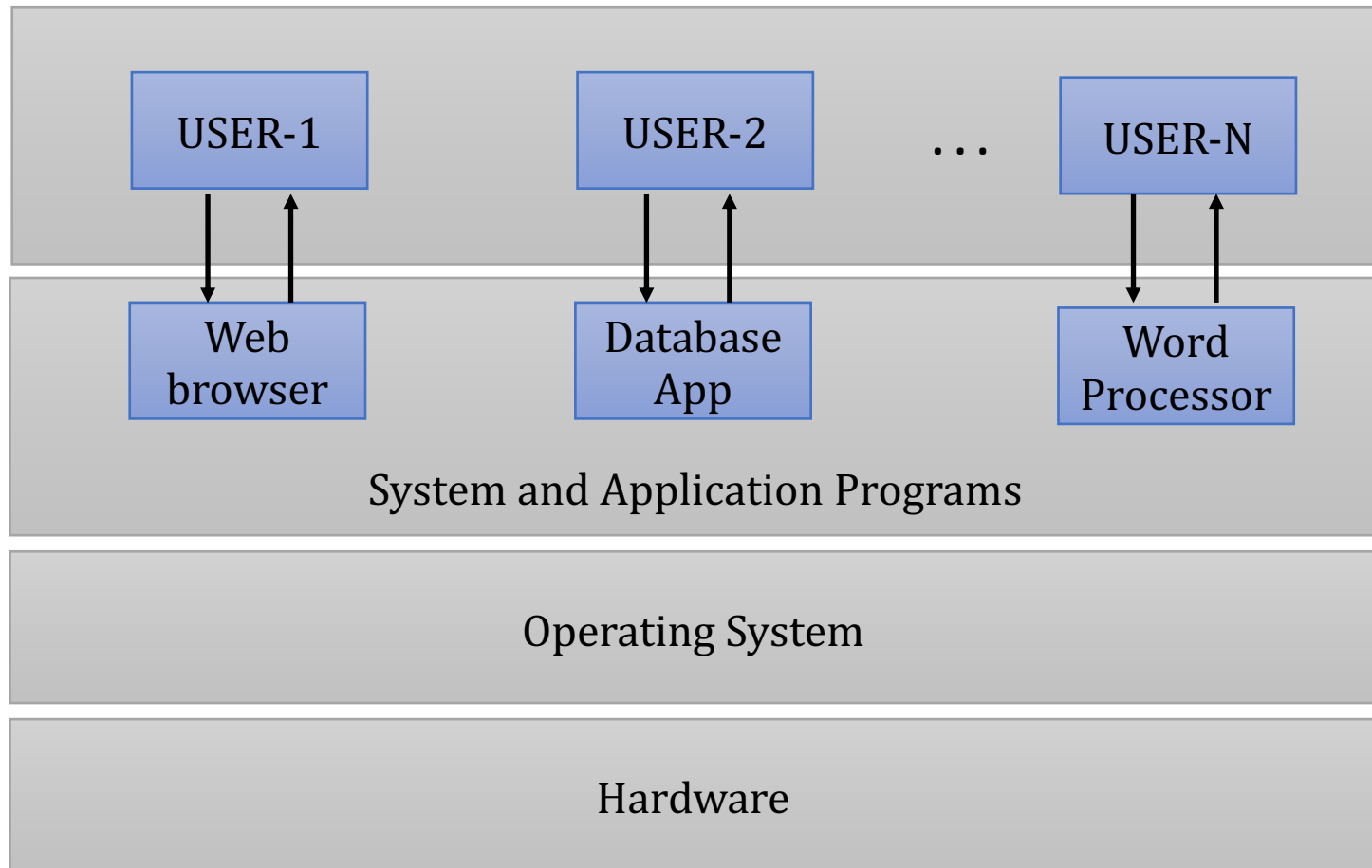


Figure: Abstract view of the components of a computer system.

Goals of the OS

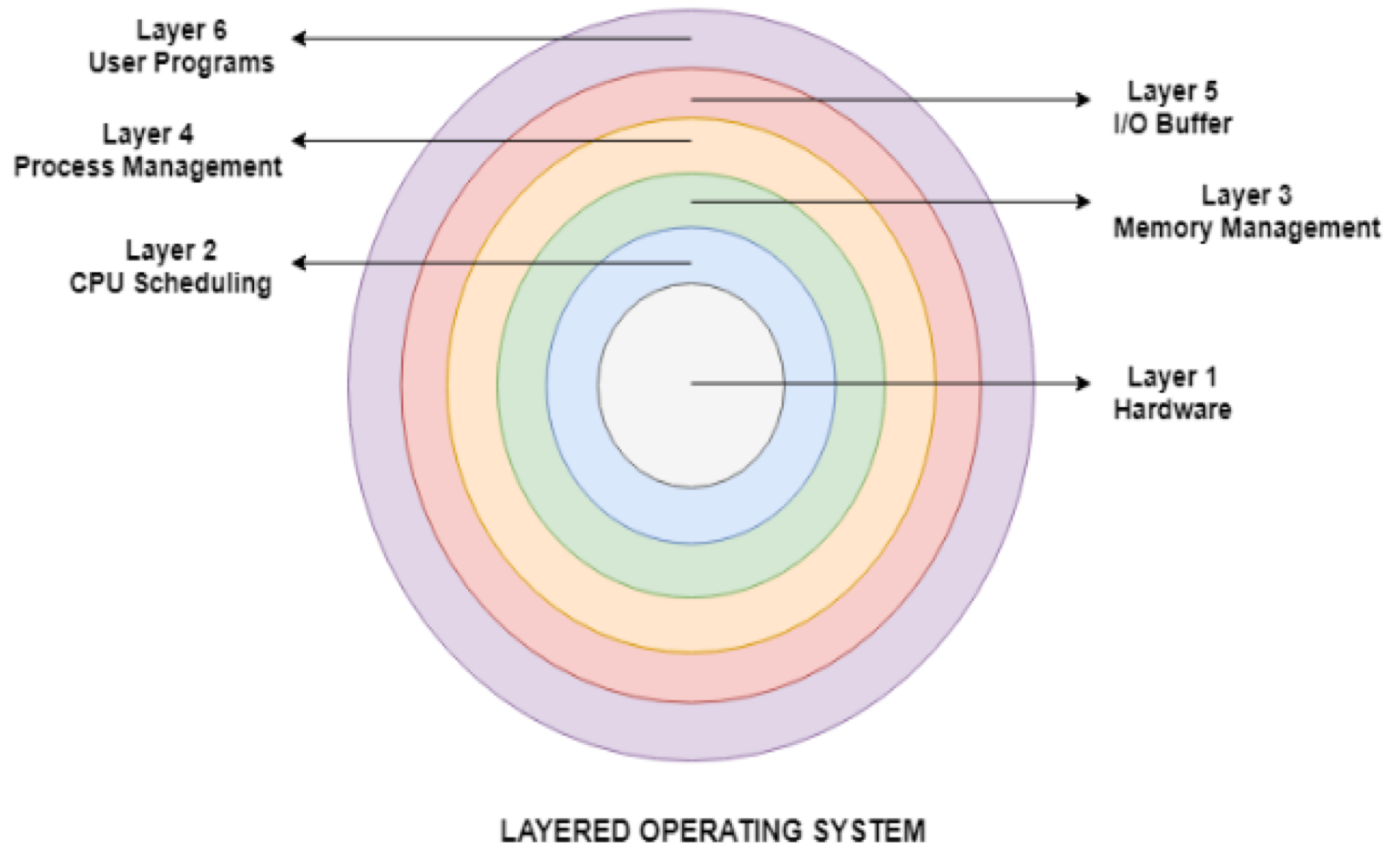
- ▶ **Primary Goal:** The primary goal of an Operating System is to provide a ***user-friendly and convenient environment.***
- ▶ **Secondary Goal:** The secondary goal of an Operating System is ***efficiency.***

Functions of OS

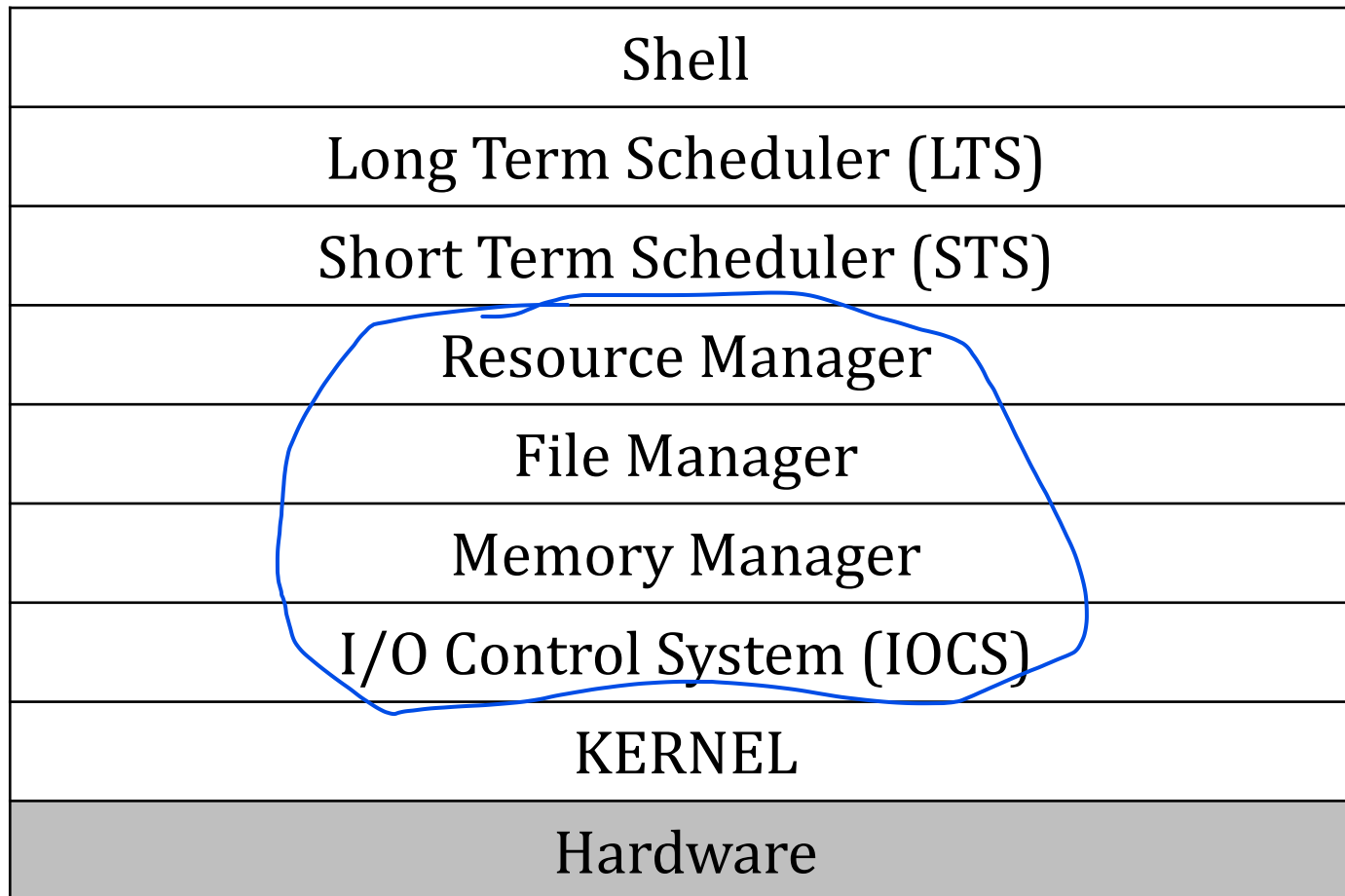
FW, FNVs

- ▶ **Process Management**
- ▶ **Memory Management**
- ▶ **I/O Device Management**
- ▶ **File Management**
- ▶ **Network Management**
- ▶ **Virtual Memory**
- ▶ **Security & Protection**

OS: Layered Structure



OS: Layered Structure

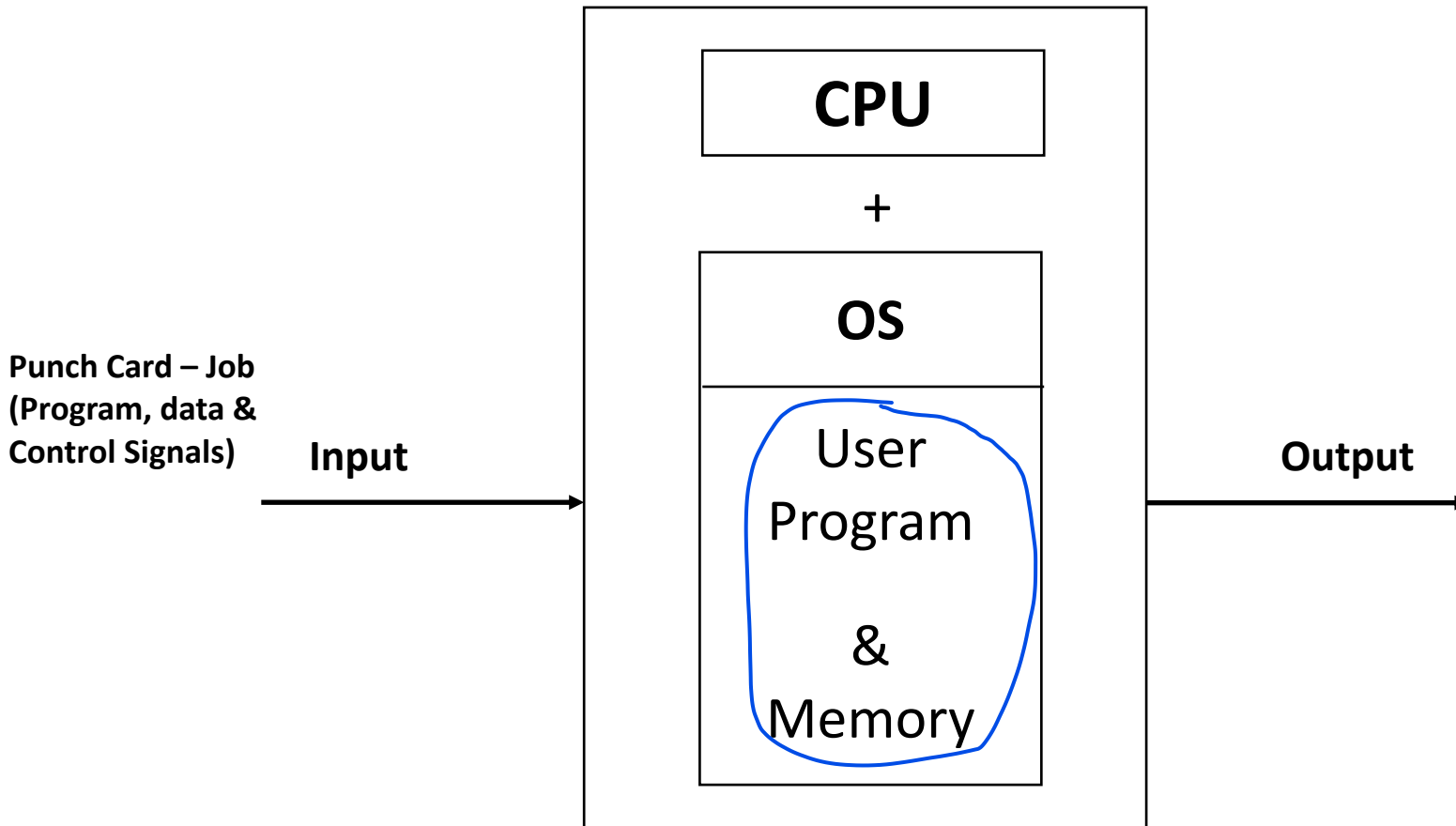


Types of OS

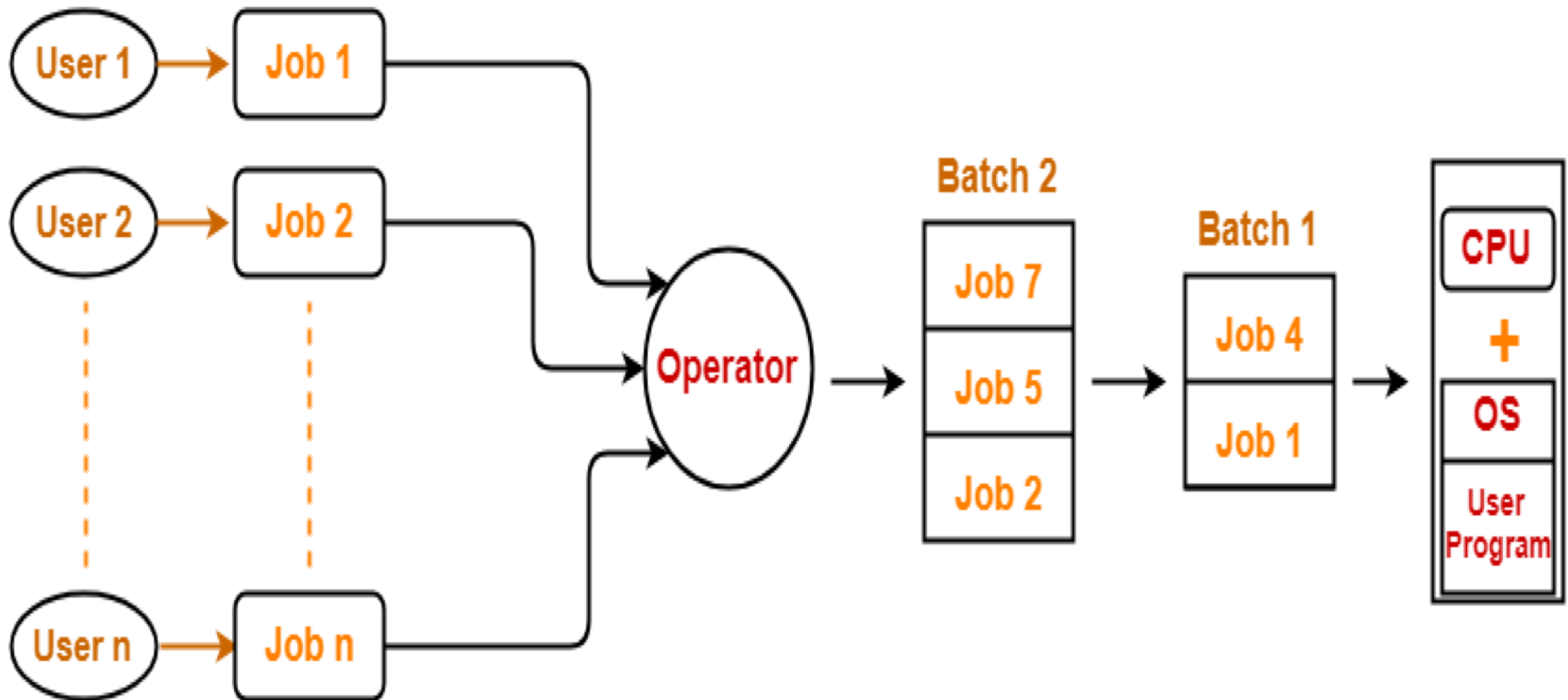
- ▶ Batch OS
- ▶ Multi Programming OS
- ▶ Multi Processing OS
- ▶ Time Sharing OS
- ▶ Multi Threading OS
- ▶ Real Time OS

Problem before Batch OS

Computer



Batch Operating System



Batch Operating System

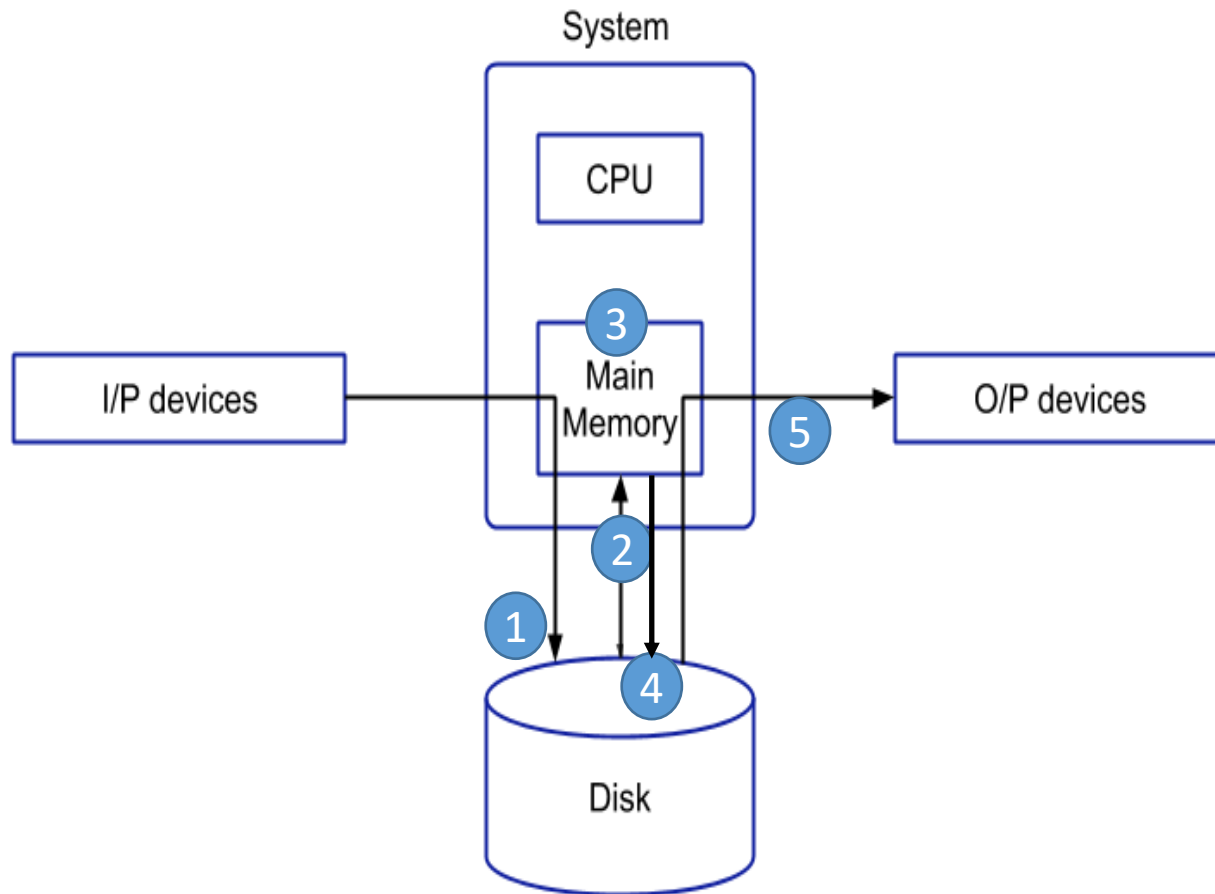
Advantages

- ▶ *Time saving* from activities like **loading compilers**.
- ▶ **No manual intervention** is needed during the execution of a batch.
- ▶ Processors of the batch systems know **how long the job** would be in the system when it is in queue.
- ▶ **Multiple users** can share the batch systems.
- ▶ The **idle time** for batch system is **very less**.
- ▶ It is easy to manage *large work* repeatedly in batch systems.

Disadvantages

- ▶ *Memory limitation.*
- ▶ The *computer operators* should be well known with batch systems.
- ▶ Batch systems are *hard to debug.*
- ▶ It is sometime *costly.*
- ▶ The *other jobs will have to wait for an unknown time* if *any job fails.*

Spooling (Simultaneous Peripheral Operations Online)

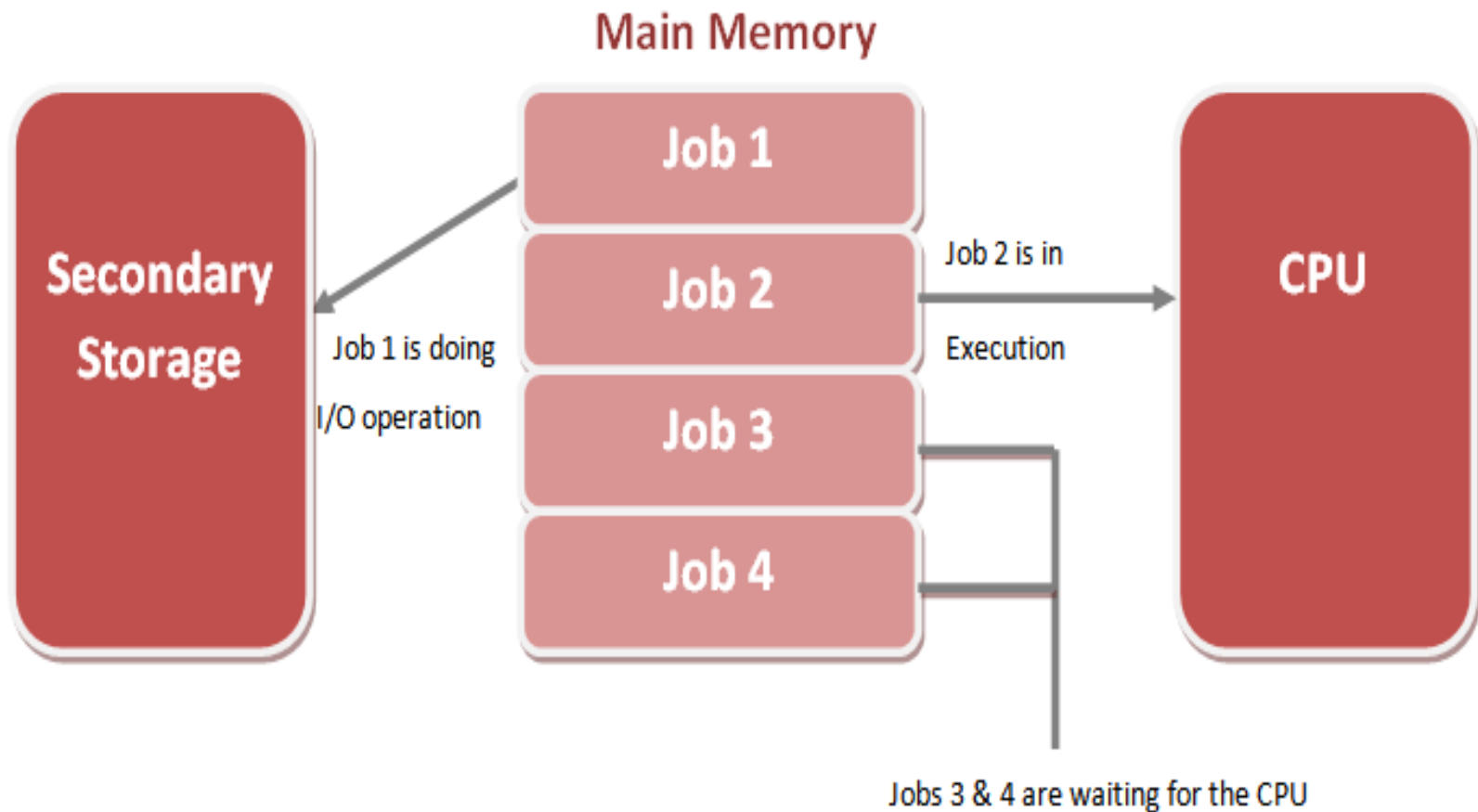


Spooling

Divide Advantages/Disadvantages

- ▶ No interaction of I/O devices with CPU, so the CPU need not wait for the I/O operation to take place.
- ▶ The CPU is kept busy most of the time and hence it is not in the idle state i.e. *CPU utilization is high*.
- ▶ More than one *I/O devices can work simultaneously*.
- ▶ **Disadvantages:**
 - ▶ At Starting spooling was uniprogramming.
 - Next process need to wait even if first is waiting for I/o or other command.

Multiprogramming



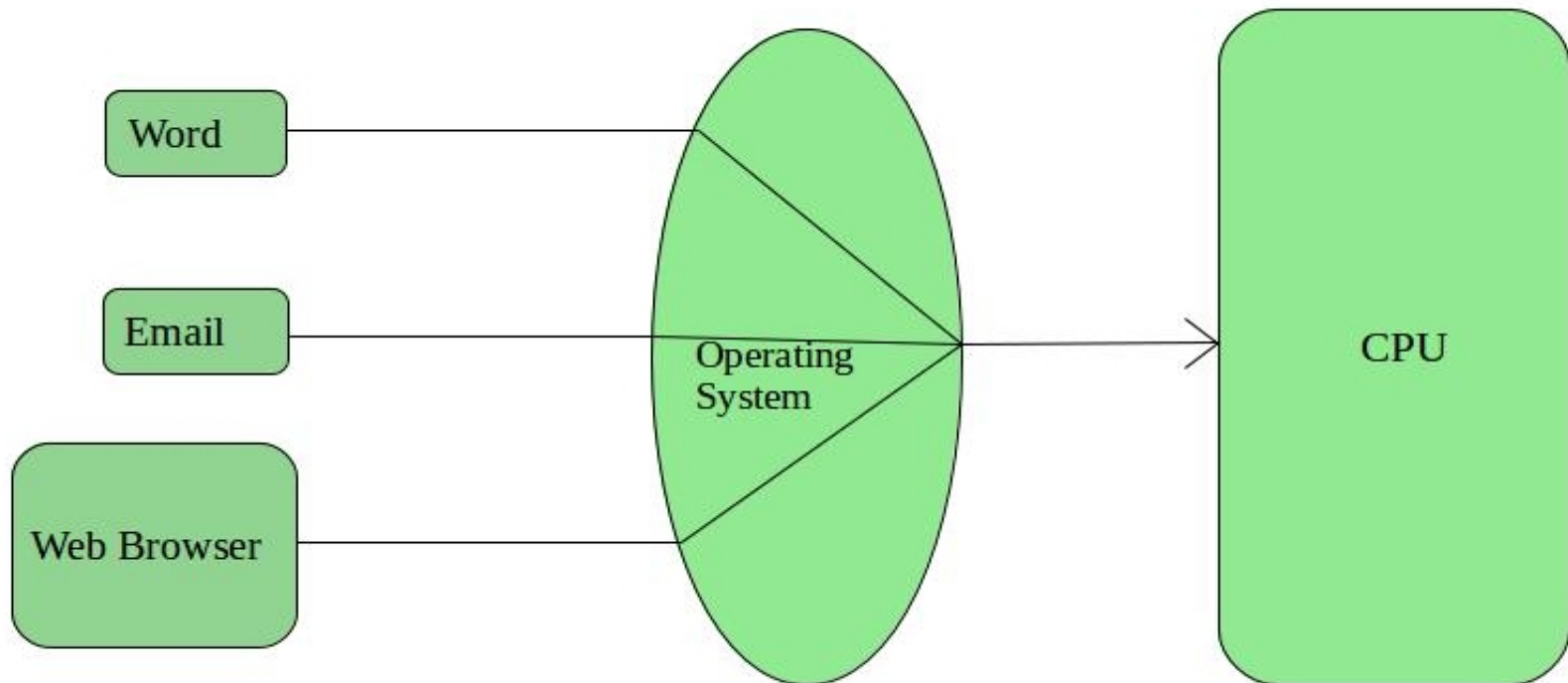
Advantages

- ▶ High and efficient *CPU utilization*.
- ▶ *User feels* that many programs are allotted CPU almost simultaneously.
- ▶ *Less waiting time, response time.*
- ▶ May be extended to *multiple users* (servers).
- ▶ *Useful when load is more.*

Disadvantages

- ▶ Difficult *CPU* or *Process scheduling*.
- ▶ To accommodate many jobs in memory, *memory management is required*.
- ▶ *Memory fragmentation*.
- ▶ *Paging*.

Multitasking OS/Time sharing/Fair share/ Multiprogramming with Round Robin



Multitasking OS

Advantages

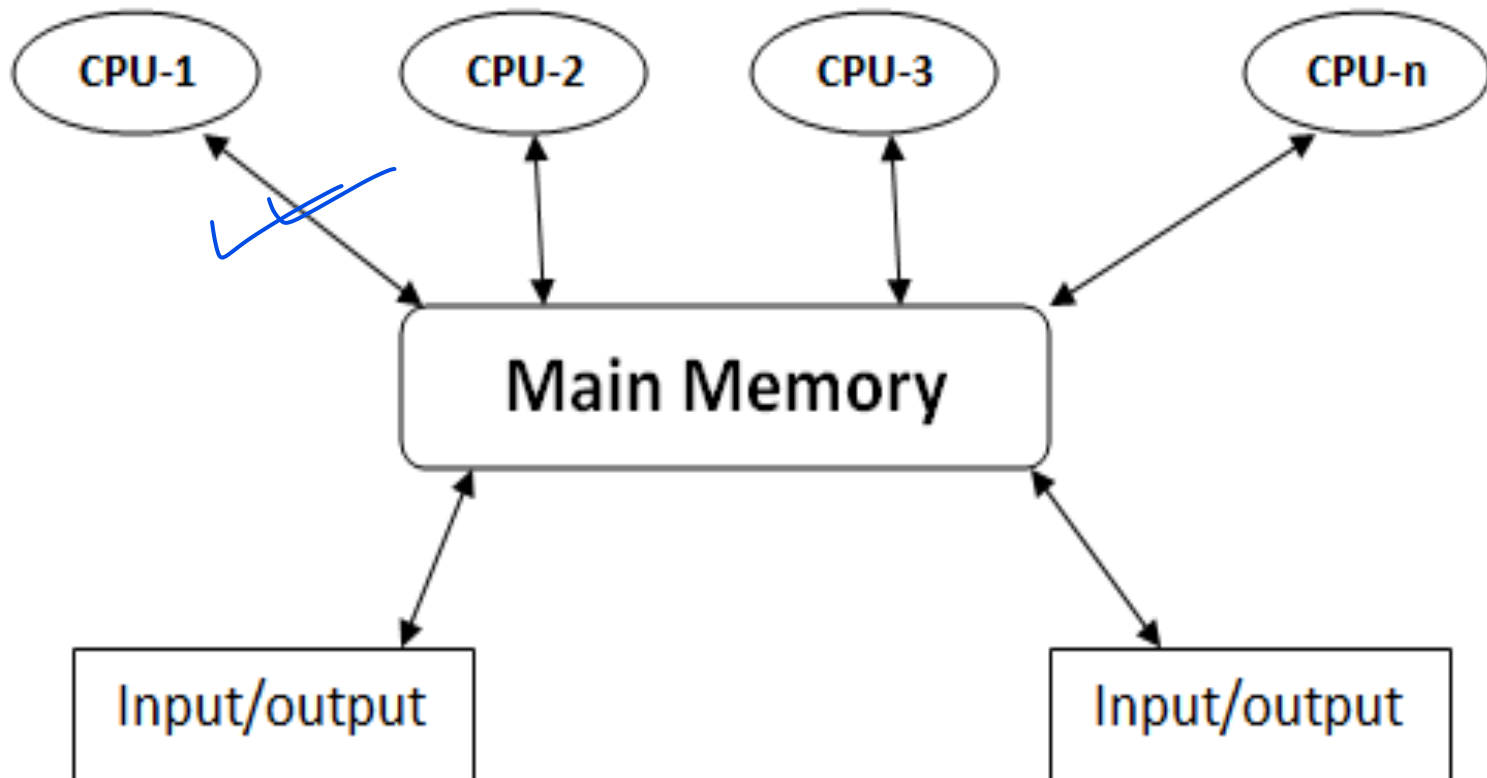
- ▶ Multitasking is *multiprogramming with time-sharing*.
- ▶ Provides the advantage of quick response.
- ▶ Avoids duplication of software.
- ▶ Reduces CPU idle time.



Disadvantages

- ▶ Problem of reliability.
- ▶ Question of security and integrity of user programs and data.
- ▶ Problem of data communication.

Multiprocessing



Multiprocessing

Types of Multiprocessing

Two types of multiprocessing:

▶ Symmetric: One OS controls all the CPUs.

Each CPU has equal rights.

▶ No Master, No Slave

▶ Asymmetric: *System task* on one processor and *Application* on the other.

▶ Master, Slave

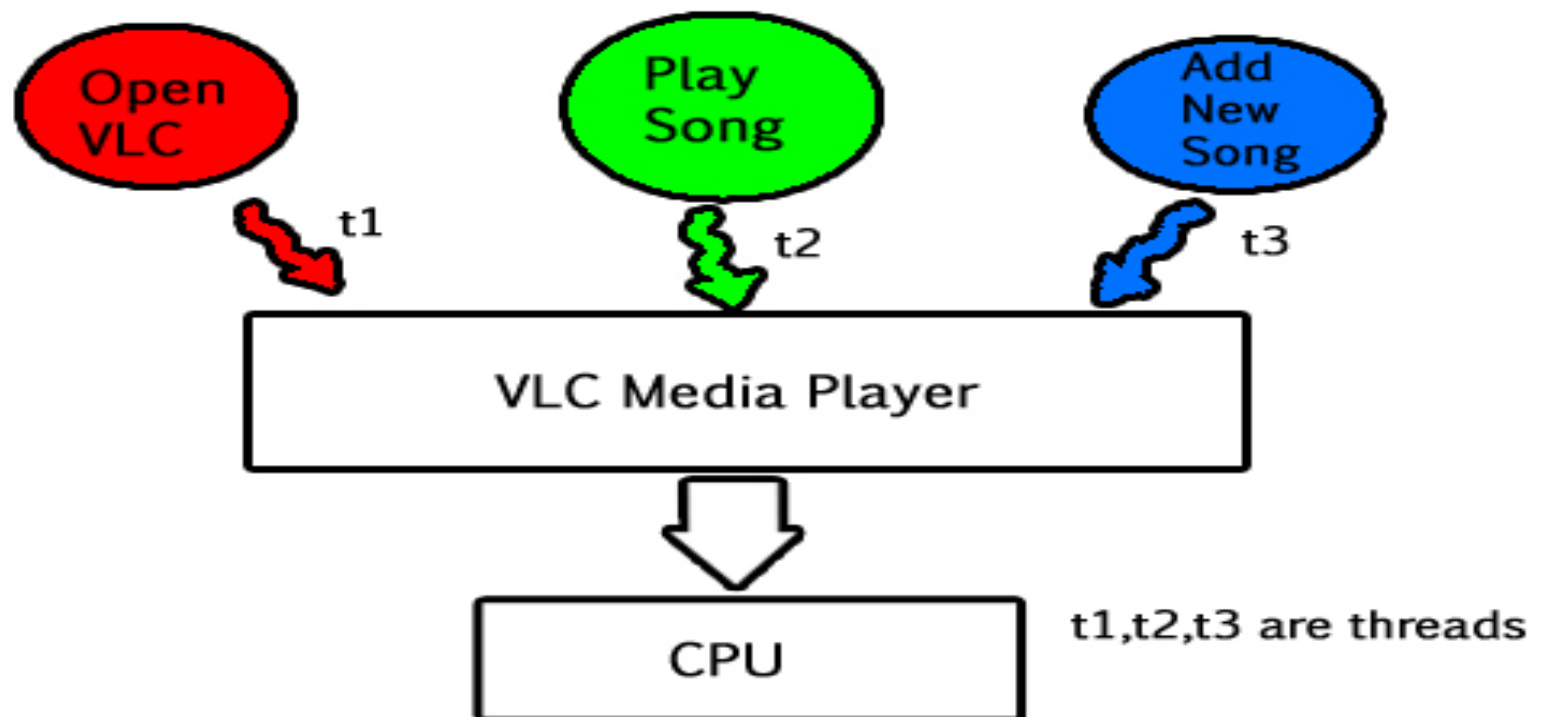
Advantages

- ▶ *Increased Throughput* due to multiple CPUs
- ▶ *Increased Reliability* – one fails: no issue
- ▶ *Cost Saving*: separate memory etc.
- ▶ *Battery efficient*: Less components.
- ▶ *True Parallel Processing*.

Disadvantages

- ▶ More complex – *Scheduling, synchronization.*
- ▶ Overhead or *coupling reduces throughput.*
 - ▶ 4 CPUs doesn't mean the 4 times of Throughput?
- ▶ *Large main memory* required.

Multi-threading



Ex: Web Server

Advantages

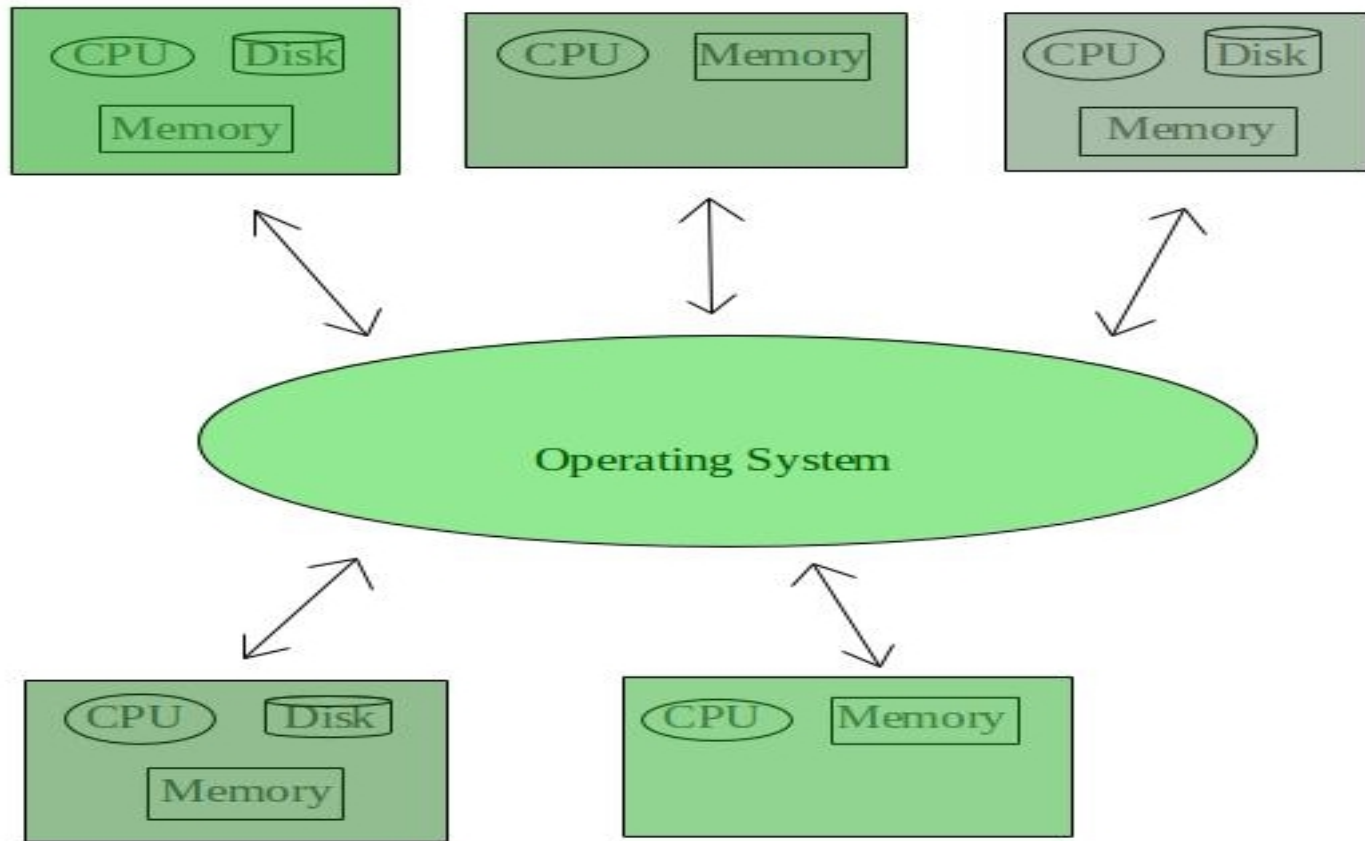
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- ▶ *Max utilisation of CPU* is possible.
- ▶ *Time Saving*.
- ▶ *Increased responsiveness:* one thread is blocked still other will continue.
- ▶ Improves the performances of *complex applications*. (Mathematical Computation)
- ▶ *Parallelizes tasks*.

Disadvantages

- ▶ Complex *debugging* and *testing* process.
- ▶ *Increased complexity* level in writing a program.
- ▶ *Unpredictable results*.

Distributed/**Loosely coupled** Systems



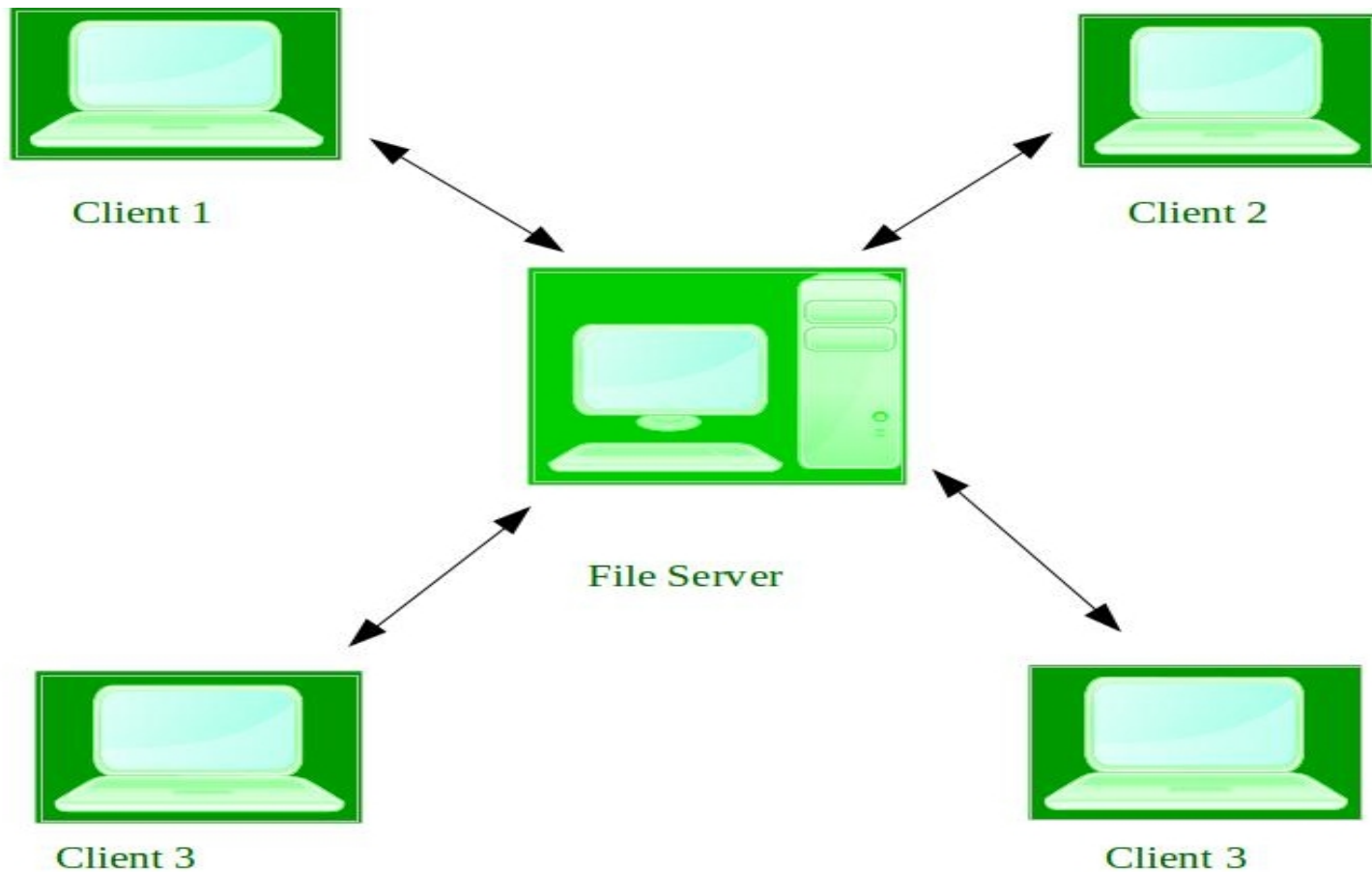
Advantages

- ▶ *Failure of one will not affect the other network communication*, as all systems are independent from each other.
- ▶ Electronic mail increases the data exchange *speed*.
- ▶ Since resources are being shared, *computation is highly fast and durable*.
- ▶ *Load on host computer reduces*.
- ▶ These systems are easily *scalable* as many systems can be easily added to the network.
- ▶ *Delay in data processing reduces*.

Disadvantages

- ▶ *Failure of the main network* will stop the entire communication
- ▶ To establish distributed systems the *language* which are used are not well defined yet.
- ▶ These types of systems are not readily available as they are *very expensive*. Not only that the *underlying software is highly complex* and not understood well yet.

Network/Tightly coupled System



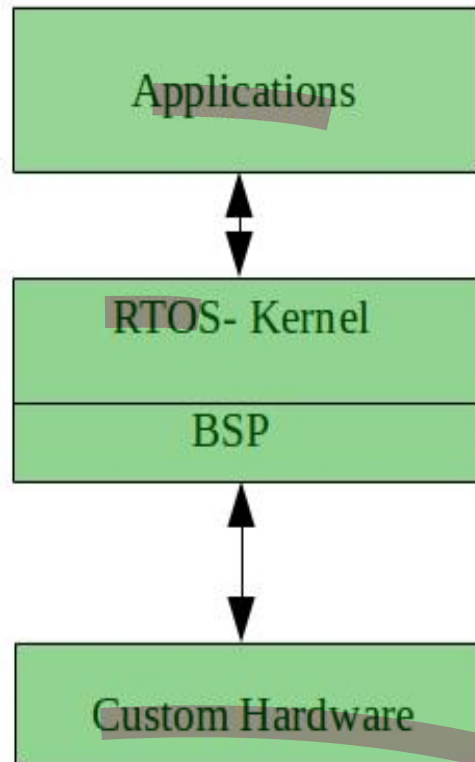
Advantages

- ▶ Highly stable *centralized servers*.
- ▶ *Security concerns* are handled through *servers*.
- ▶ New technologies and hardware *up-gradation* are easily integrated to the system
- ▶ Server access are possible remotely from different locations and types of systems.

Disadvantages

- ▶ Servers are *costly*.
- ▶ User has to depend on *central location* for most operations.
- ▶ *Maintenance* and updates are required regularly.

Real-Time Operating System



Strict time requirement: Missile system, air traffic control, robots etc.

Types of Real-Time Operating System

► Hard Real-Time Systems:

Applications where time constraints are very strict and even the shortest possible delay is not acceptable.

Ex- Automatic parachutes or air bags

Warning: Virtual memory is never used in these systems.

► Soft Real-Time Systems:

These OSs are for applications where for time-constraint is less strict.

Advantages

- ▶ **Maximum Consumption:** *Maximum utilization* of devices and system, thus more output from all the resources.
- ▶ **Task Shifting:** Time assigned for shifting tasks in these systems are *very less*. For example in older systems it takes about *10 micro seconds* in shifting one task to another and in latest systems it takes *3 micro seconds*.
- ▶ **Focus on Application:** *Focus on running applications and less importance to applications which are in queue.*

Advantages

- ▶ **Real time operating system in embedded system:** Since size of programs are small, *RTOS can also be used in embedded systems like in transport and others.*
- ▶ **Error Free:** These types of systems are *error free.*
- ▶ **Memory Allocation:** *Memory allocation is best managed in these type of systems.*

Disadvantages

- ▶ **Limited Tasks:** *Very few tasks run at the same time* and their concentration is very less on few applications to avoid errors.
- ▶ **Use heavy system resources:** Sometimes the system resources are not so good and they are *expensive* as well.
- ▶ **Complex Algorithms:** The algorithms are very *complex and difficult* for the designer to write on.

Disadvantages

- ▶ **Device driver and interrupt signals:** It needs *specific device drivers* and *interrupt signals* to response earliest to interrupts.
- ▶ **Thread Priority:** It is not good to set thread priority as *these systems are very less prone to switching tasks*.

Thank You